PRACTICE SHEET

1. The lines $(p + 2q)x + (p - 3q)y = p - q$ for different values of p and q pass through the fixed point given by which one of the following?

(a) $\left(\frac{3}{2}, \frac{5}{2}\right)$ (b) $\left(\frac{2}{5}, \frac{2}{5}\right)$

(c) $\left(\frac{3}{5}, \frac{3}{5}\right)$ (d) $\left(\frac{2}{5}, \frac{2}{5}\right)$

2. What is the angle between the two straight lines $y = (2-\sqrt{3})x + 5$ and $y = (2+\sqrt{3})x - 7$?

(a) 60° (b) 45°
(c) 30° (d) 15°

3. What is the image of the point (2, 3) in the line $y = -x$?

(a) (−3, −2) (b) (−3, 2)
(c) (−3, −2) (d) (3, 2)

4. The middle point of A(1,2) and B(x,y) is C(2, 4). If BD is perpendicular to AB such that CD = 3 unit, then what is the length BD?

(a) $2\sqrt{2}$unit (b) 2 unit
(c) 3 unit (d) $3\sqrt{2}$unit

5. If the point A (1, 2), B(2, 4) and C(3,a) are collinear, what is the length BC?

10. If p be the length of the perpendicular from the origin on the straight line $ax + by = p$ and $b = \frac{\sqrt{3}}{2}$, then what is the angle between the perpendicular and the positive direction of x – axis?

(a) 30° (b) 45°
(c) 60° (d) 90°

11. The straight line $ax + by + c = 0$ and the coordinate axes form an isosceles triangle under which one of the following conditions?

(a) |a| = |b| (b) |a| = |c|
(c) |b| = |c| (d) None of these

12. The coordinates of P and Q are (−3, 4) and (2, 1), respectively, If PQ is extended to R such that PR = 2QR, then what are the coordinates of R?

(a) (3, 7) (b) (2, 4)
(c) $\left(-\frac{1}{2}, \frac{5}{2}\right)$ (d) (7, −2)

13. Which one of the following points on the line $2x - 3y = 5$ is equidistant from (1, 2) and (3, 4)?

(a) (7, 3) (b) (4, 1)
(c) (1, −1) (d) (−2, −3)

14. The following question consists of two statements, one